THE UNIVERSITY OF BRITISH COLUMBIA

CONFINED DIFFUSION OF NANOPARTICLES USING iSCAT MICROSCOPY

by

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Abstract

The aim of this thesis is to characterize the full diffusion model of complex nanoparticles. To accomplish this, the 3D position of a nanoparticle(20-100 nm) must be measureable with a time resolution on the other of the diffusion length (ms). Interferometric light scattering microscopy (iSCAT) is used in tandem with convex-lens induced confinement (CLiC) to simultaneously confine particles, while obtaining a spatial signal.

This thesis developed a novel method of measuring axial position using iSCAT data. The method was applied to a hard-wall axial confinement potential. The system's ionic conditions, particle charge, and confinement length were varied to characterize the effect of the various variables.

The application of this thesis was directly applied to a more common, commericially assembled fluorescent microscope. The applications of the iScat confinement measurements allow for a direct correction to fluorescet microscopy calculations of diffusion in confinemnt. To accomplish this, an off-the-shelf optical system was designed. The system consists of a 3-color laser excitation with alternating laser excitation capabilities. The Nikon Ti2-Eclipse microscope chassis was used for this system in tandem with a 2-camera twincam system for dual color imaging. Imaging of nanoparticles was performed under various confinement regimes and corrections to the measuements was performed.

The results showed...

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test [1]

Chapter 1

Introduction

1.1 Movivation

- What is the major field i am in? - what are the biological questions - how are they typically slved

- problems with the current solutions

1.2 Outline for Thesis

- Chapter 1 is an introduction/background - Chpater 2 discusses application of Widefield iSCAT for SPT in the Rayleigh Regime - Conclusion

Chapter 2

SPT in confinement principles

2.1 Brownian Motion

- Introduce the concept of Brownian Motion - Histroy of brownian Motion - applications of simple

- discuss an outlook of what has been done since then - outline of this section

2.1.1 Langevin Dynamics

- Brownian motion can alternatively be described using a Langevin Model. - introduce langevin equation - velocity correlation - Arrive at the Fluc-Dissipation - relate it to Einstein's formulation and show that given a stokes drag (sphere)

$$D = \frac{k_b T}{\gamma_0} \tag{2.1}$$

2.1.2 Smoluchowski Equation

- Since Langevin is kind of convient, how do we acctualy link the two methods?

- Fokker plank introdution motivated from the master equation - - ideally summarize van kampen.

- arrive at the general Smoluchowski equation for a random walker - simple plots for descriptive purposes.

2.2 Confined Diffusion Processes

Although diffusion is interesting, there is a wealth of complexity in confinement due to Hydrodynamic effects - quote 111 years here

- outline of this seciton

2.2.1 Hydrodynamic Confinement

- confinemnt from Hydrodynamics - frictional slowing - ...

2.2.2 Geometric Confinement

- i need to check references on if this is relate

2.2.3 Electrostatic Potentials

2.2.4 Example: Polystyrene Bead

2.3 In-Vivo Considerations

- Now that we more or less know how BM works, why is it important in cells?

2.3.1 why does this matter for biological applications

2.3.2 Examples of Brownian Motion in Biology

2.4 Brownian motion as a method for measurements of size in biology

2.4.1 survey of methods using in this field

- Optical Methods - Bulk: DLS, - Single Molecule: SPT, NTA, FCS,

2.4.2 how confinemen effects these measurements

- Literature on why these methods are lacking

2.4.3 Probing of surface interactions with these methods

Chapter 3

Widefield backscattering iscat for SPT in 3d confinement tracking and analysis

3.1 Questions

3.2 Rayleigh scattering of nanoparticles

3.2.1 Parameterization used in this thesis

3.3 Methods

3.3.1 Optical System

3.3.2 Convex-lens induced confinement

3.4 Results i guess

3.5 Discussion of Results

Chapter 4

Conclusion

4.1 Conclusion

4.2 Future outlook

Bibliography

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